MILITARY SPECIFICATION SHEET

ELECTRON TUBES, POWER

TYPES 7457, 7842

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tubes described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Tetrode, UHF, Ceramic-metal, ruggedized, Fl = 1,215 MHz

Outlines: See figures 1 and 2

Mounting position: Any

Weight: 2 ounces nominal

ABSOLUTE MAXIMUM RATINGS:

Parameter: / Units:	Ef V	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ib mAdc	Ic1 mAdc	Pg2 W	Pi W	Pp W
AB Audio:	6.3 ±10%	1,000	---	300	180	30	4.5	180	115
AB SSBSC:	6.3 ±10%	1,000	---	300	250 Note 2	30	4.5	180	115
C Telep:	6.3 ±10%	800	-100	300	150	30	3.0	120	75
C Teleg:	6.3 ±10%	1,000	-100	300	180	30	4.5	180	115
TEST CONDITION:	6.3	1,000	Adj.	300	---	--	---	---	---

ABSOLUTE MAXIMUM RATINGS:

Parameter: / Units:	Anode Core & Seal T °C	tk sec (min)	Alt. ft	Cooling Note 1
AB Audio:	250	60	40,000	---
AB SSBSC:	250	60	40,000	---
C Telep:	250	60	40,000	---
C Teleg:	250	60	40,000	---
TEST CONDITION:	---	120	---	Note 3

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	12					
1301	Heater current	---		If	2.90	3.55	A
1266	Total grid current	4	Ec1/Ib = 115 mAdc	Ic1	----	-20	µAdc
1261	Electrode voltage (1) (grid)	---	Ec1/Ib = 115 mAdc	Ec1	-6.0	-18.0	Vdc
1261	Electrode voltage (2) (grid)	---	Ec1/Ib = 1.0 mAdc	Ec1	----	-48	Vdc
1256	Electrode current (screen)	---	Ec1/Ib = 115 mAdc	Ic2	-8.0	+2.0	mAdc
1231	Pulsing emission sinusoid	5	eb=ec2=ec1 = etd; etd/ik = 10 a.	etd	----	300	v
	Quality conformance inspection, part 2	11					
1266	Primary grid emission (control)	---	Eg1/Pg1 = 2.0 W; t = 30; a and g2 grounded	Isg1	----	-12	µAdc
1266	Primary grid emission (screen)	---	Eg2/Pg2 = 4.5 W; t = 30; a and g1 grounded	Isg2	----	-25	µAdc
1366	Interelement leakage resistance, cold	6	Test E = 200 Vdc	R	1.0	----	Meg
2214	RF useful power output	7	Ef = 5.7 V; t = 120; F = 400 ±20 MHz; Ib = 180 mAdc max; Ic2 = +15 mAdc max; Ic1 = 30 mAdc max; Pd = 3.3 W max	Po	80	----	W (useful)
1236	Power oscillation	---	F = 15 MHz; t = 120; Eb = 850 Vdc; Ec2 = 300 Vdc max; Ib = 150 mAdc; Rg1 = 4,000 ohms max; Ic1 = 30 mAdc; Pg2 = 4.5 W (max)	Po	80	----	W (useful)
1331	Direct-interelectrode capacitance	---	Capacitance fixture per 289-JAN	Cg1k Cg1g2 Cg2p Cg1p Cpk Cg2k	11.8 17.3 4.0 ---- ---- ----	15.2 21.9 5.1 0.065 0.019 1.30	pF pF pF pF pF pF
	Quality conformance inspection, part 3	11					
1506	Heater-cycling life	---	Group C; Ef = 8.5 V; t = 150 "ON", 300 "OFF"	----	384	----	cycles
----	Heater-cycling life end point:						
1301	Heater current	---		If	2.70	3.55	A

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued	11					
----	Life test	4, 7	Group C; Ef = 5.7; F = 400 ±20 MHz; Ib = 180 mAdc; Icl = 30 mAdc max; Ecl = value obtained for Ecl in electrode voltage (2) test per part 1 herein	t	500	---	hrs
----	Life test end points:						
1231	Pulsing emission sinusoid		etd/ik = 7.5 a	etd	---	400	V
1266	Primary grid emission (control)			Isg1	---	-12	µAdc
1266	Primary grid emission (screen)			Isg2	---	-25	µAdc
2214	RF useful power output			Po	65	---	W (useful)
1042	Shock, specified pulse (long duration)	8	Test condition A; Ef only	---	---	---	---
----	Vibration (noise)	8, 10	Ebb = 300 Vdc; Ec2 = 250 Vdc; Ecl/Ib = 10 mAdc; Rp = 2,000 ohms	Ep	---	15	Vac
----	Shock and vibration (noise) end points:						
1266	Total grid current			Icl	---	-20	µAdc
1261	Electrode voltage (1)			Ecl	-6.0	-18.0	Vdc
1002	Barometric pressure, reduced	8	Pressure = 141 mmHg; 1,300 Vdc or 920 Vac applied between a and k, with k, g1, g2 tied together; Ef = 0	---	---	---	---

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NOTES:

1. Type 7457:

   Sufficient airflow shall be supplied to limit the anode core and all seal temperatures to 250°C maximum.

   Type 7842:

   The anode conduction cylinder on the tube under test (TUT) must be thermally coupled to a suitable heat sink to limit the temperature of the anode core to the maximum value of 250°C. The respective seal temperatures of the anode, control grid, screen grid, cathode, and heater terminals shall also be limited in regard to this maximum temperature value by equal thermal coupling to the heat sink or other means if required.

   With either tube type, where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

2. The maximum dc anode current (Ib) at the peak of the current curve is 250 mAdc for a signal having a minimum peak-to-average power ratio of 2.0. The maximum rating for a signal having a minimum peak-to-average power ratio of less than 2.0, such as is obtained in single-tone operation, is 180 mAdc. During short periods of circuit adjustment under single-tone conditions, the average anode current may also reach the level of 250 mAdc.

3. During all electrical tests involving application of heater voltage, sufficient cooling may be supplied for the anode and seals to maintain temperatures within the maximum rating.

4. Cooling shall be set up to maintain the tube temperature at the hottest point on the tube envelope at 225 ±25°C. This set up should be permanent and need be verified only as often as required to maintain equipment accuracy.

5. With anode grid No. 1 and grid No. 2 tied together, apply a nominal 1/2 sinusoidal pulse voltage source between plate and cathode. The applied voltage source shall meet the following requirements: Pulse duration = 2 µs minimum and not greater than 10 µs at the base line of the pulse, prr = 60 pps minimum. Adjust the pulse amplitude until the specified peak cathode current (ik) is obtained. Measurement shall be made at the end of one minute or when stability is reached, whichever is first.

6. Prior to performing this test, the TUT shall be cooled at room ambient temperature for 30 minutes. Using the test circuit shown on the figure of Method 1366, or equivalent, measure the resistance in both directions, between any two adjacent electrodes, except across the heater terminals. If the resistance value measured is below that required, repeat this test after a 10-minute interval. The TUT shall be rejected if it fails again.

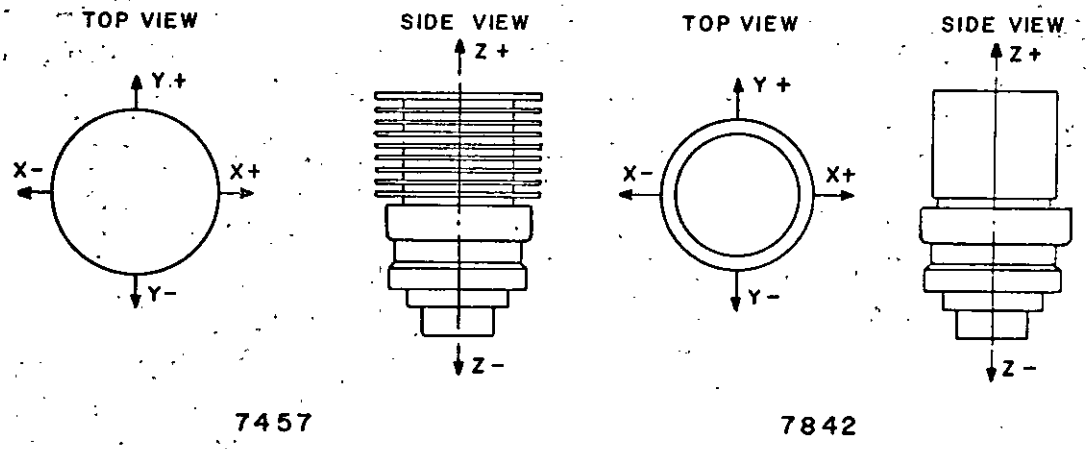
7. The TUT shall be tested in a grid-driven amplifier circuit. Adjust the Ecl bias supply and tune the circuit for maximum useful power output. The specified driving power (Pd) shall be measured with a Bird Thru-line Wattmeter, or equivalent. Driving power = forward power minus reflected power. The grid voltage supply shall have an effective impedance of 500 ohms maximum.

8. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test shall be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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NOTES: -Continued

9. Reference planes for environmental tests are as follows:



7457 7842

10. The TUT shall be vibrated in the X, Y, and Z planes under the following conditions:

Frequency (Hz)	Double Amplitude (inches)	Acceleration (G)	Approx. Sweep Duration (minutes)
5-10	0.080 ±10%	---	1
10-15	---	0.41 ±10%	.5
15-75	0.036 ±10%	---	.5
75-2000	---	10 ±10%	6

Cycling test: The frequency shall be swept from 5 to 2,000 Hz with approximately logarithmic progression, and shall require approximately 8 minutes to traverse the range. This constitutes one cycle. The TUT shall be vibrated for one such cycle in each of the tnree planes.

Record maximum vibrational noise output from 5-75 Hz using the Bruel and Kjaer type 2409 true rms meter, or equivalent. Record vibrational noise output response curve from 75-2000 Hz using the Bruel and Kjaer automatic recorder, or equivalent.

11. When the listed conditions and requirements are the same, and the tubes have been manufactured in the same production period, the listed tests may be performed on any of the types of the same basic 7457 or 7842 construction and satisfy the requirements listed herein.

12. The AQL for all tests listed under quality conformance inspection, part 1, shall be 0.65, inspection level II.

13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Review activity:
Air Force - 99

User activity:
Air Force - 11

Preparing activity:
Air Force - 85

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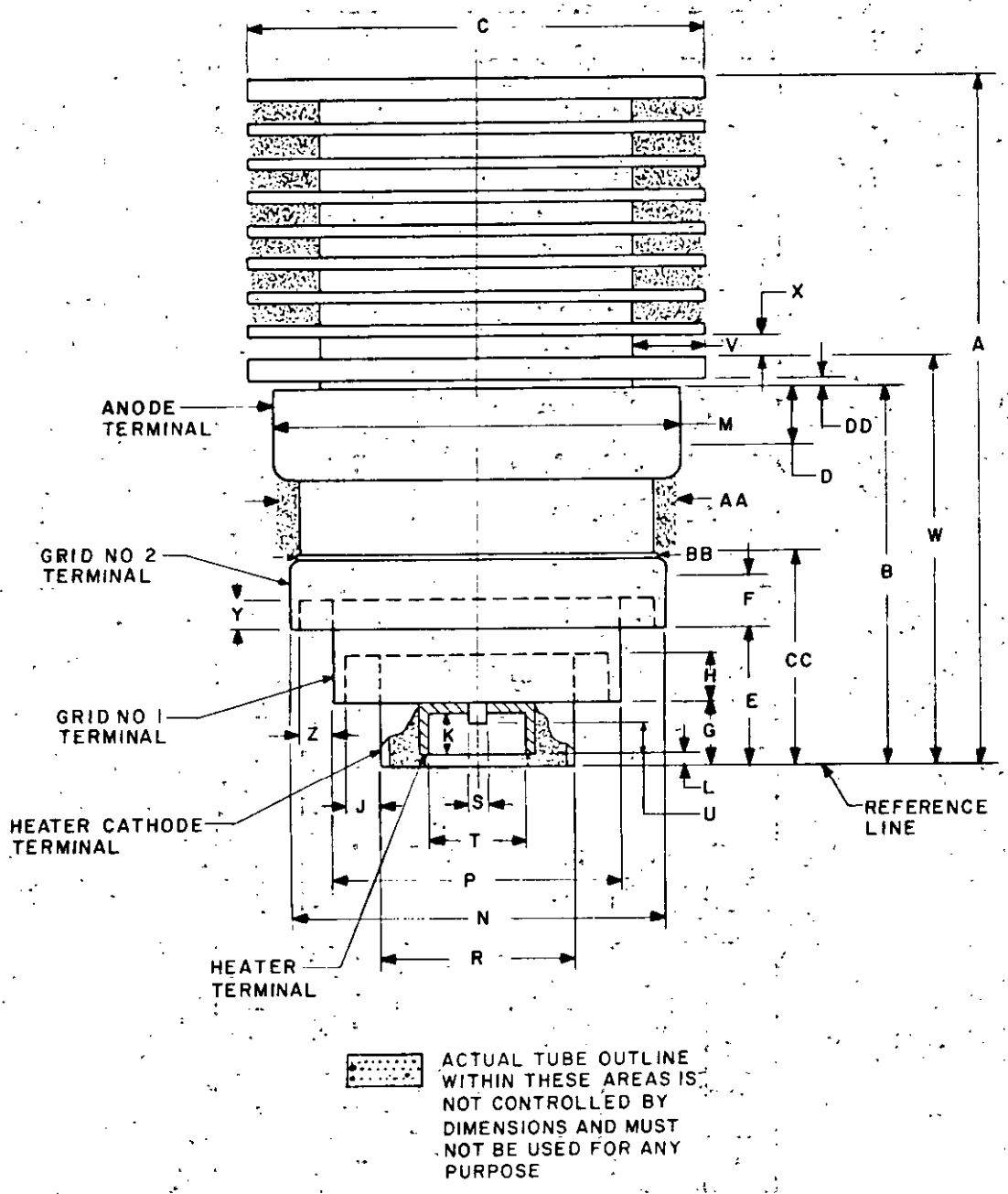


FIGURE 1. Outline drawing of electron tube type 7457.

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
Quality conformance inspect., part 1 Gage check					a,s
Quality conformance inspect., part 2					
A	1.805	1.955	45.85	49.66	
C	1.235	1.265	31.37	32.13	
Quality conformance inspect., part 3					m
B	.990	1.080	25.15	27.43	
E	.340	.410	8.64	10.41	
G	.150	.200	3.81	5.08	
M	1.085	---	27.56	---	e
N	.985	---	25.02	---	f
P	.735	---	18.67	---	g
R	.480	---	12.19	---	h
T	---	.260	---	6.60	
W	1.090	1.180	27.69	29.97	
V	.200	---	5.08	---	
X	.050	---	1.27	---	q
Quality conformance inspect., part 3					r
D	.165	---	4.19	---	
F	.140	---	3.56	---	
H	.120	---	3.05	---	b
J	.095	---	2.41	---	b
K	.100	---	2.54	---	d
L	---	.050	---	1.27	
S	---	.072	---	1.83	
U	.054	---	1.37	---	d
Y	.060	---	1.52	---	c
Z	.090	---	2.29	---	c
BB	---	---	---	---	k
CC	.600	---	15.24	---	
DD	.035	---	.89	---	
Reference					
AA	---	---	---	---	j

TUBE TYPE 7457 DIMENSIONS

FIGURE 1. Outline drawing of electron tube type 7457 - Continued.

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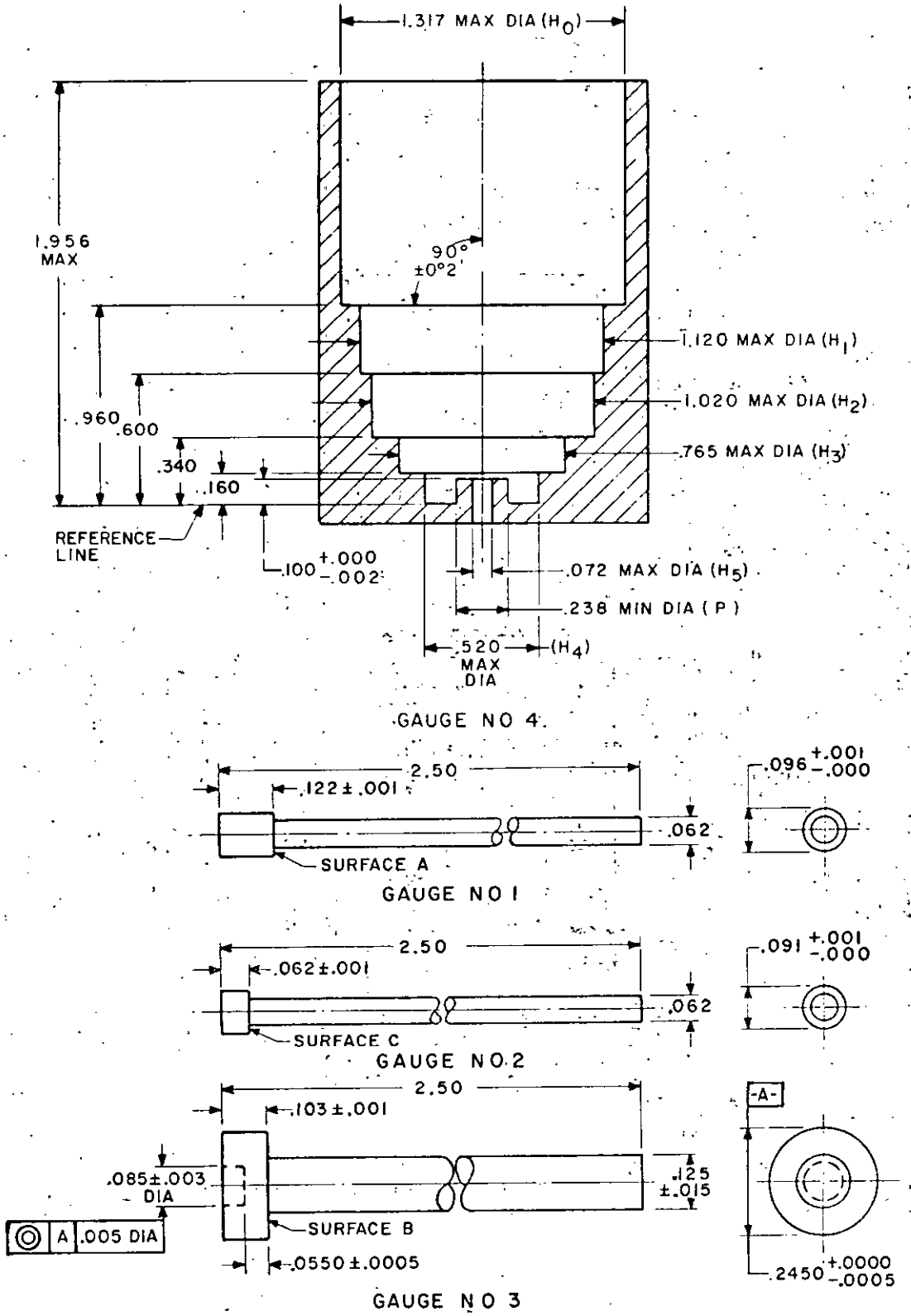


FIGURE 1. Outline drawing of electron tube type 7457 - Continued.

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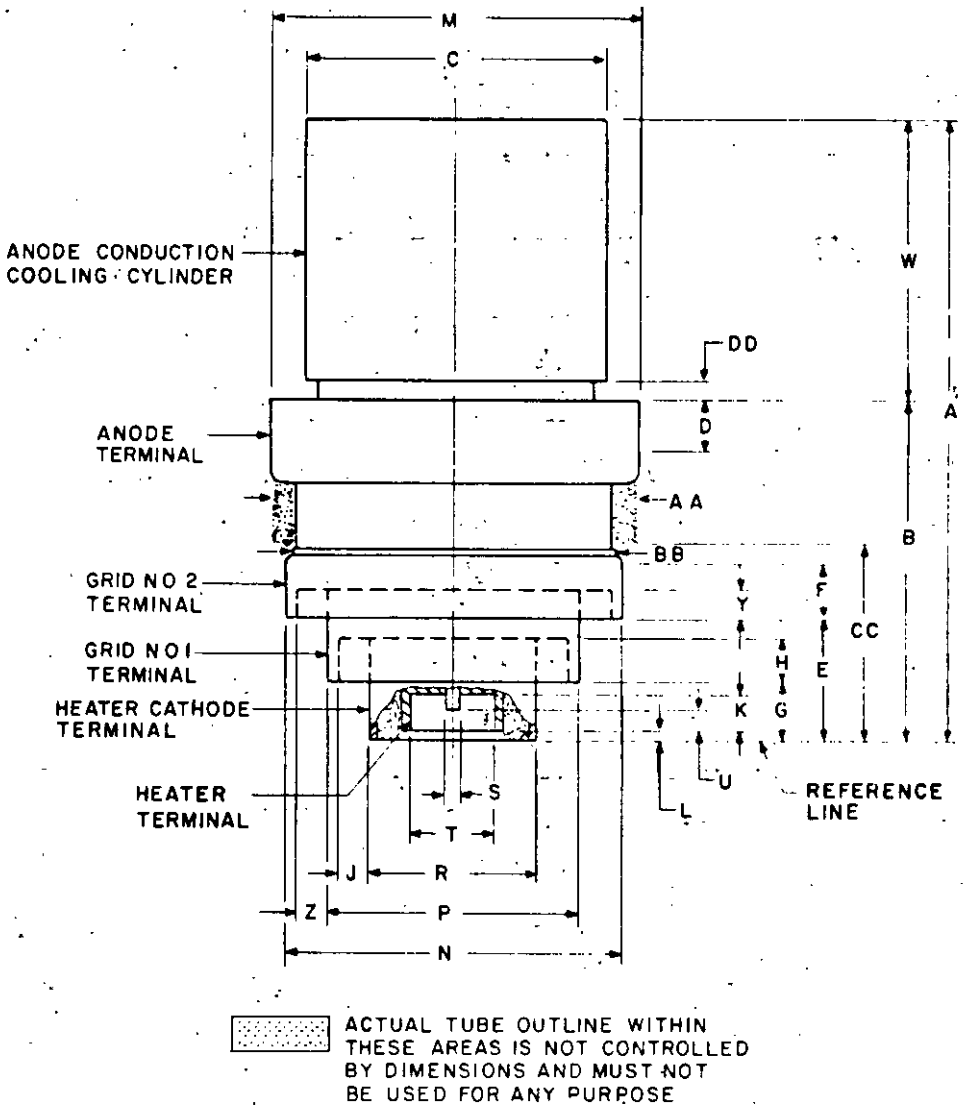


FIGURE 2. Outline drawing of electron tube type 7842.

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
Quality conformance inspect., part 1 Gage check					a,s
Quality conformance inspect., part 2					
A	1.805	1.955	45.85	49.66	
C	.895	.905	22.73	22.99	
Quality conformance inspect., part 3					m
B	.990	1.080	25.15	27.43	
E	.340	.410	8.64	10.41	
G	.150	.200	3.81	5.08	
M	1.085	---	27.56	---	e
N	.985	---	25.02	---	f
P	.735	---	18.67	---	g
R	.480	---	12.19	---	h
T	---	.260	---	6.60	
W	.780	---	19.81	---	
Quality conformance inspect., part 3					r
D	.165	---	4.19	---	
F	.140	---	3.56	---	
H	.120	---	3.05	---	b
J	.095	---	2.41	---	b
K	.100	---	2.54	---	d
L	---	.050	---	1.27	
S	---	.072	---	1.83	
U	.054	---	1.37	---	d
Y	.060	---	1.52	---	c
Z	.090	---	2.29	---	c
BB	---	---	---	---	k
CC	.600	---	15.24	---	
DD	.035	---	.89	---	
Reference					
AA	---	---	---	---	j

TUBE TYPE 7842 DIMENSIONS

FIGURE 2. Outline drawing of electron tube type 7842 - Continued.

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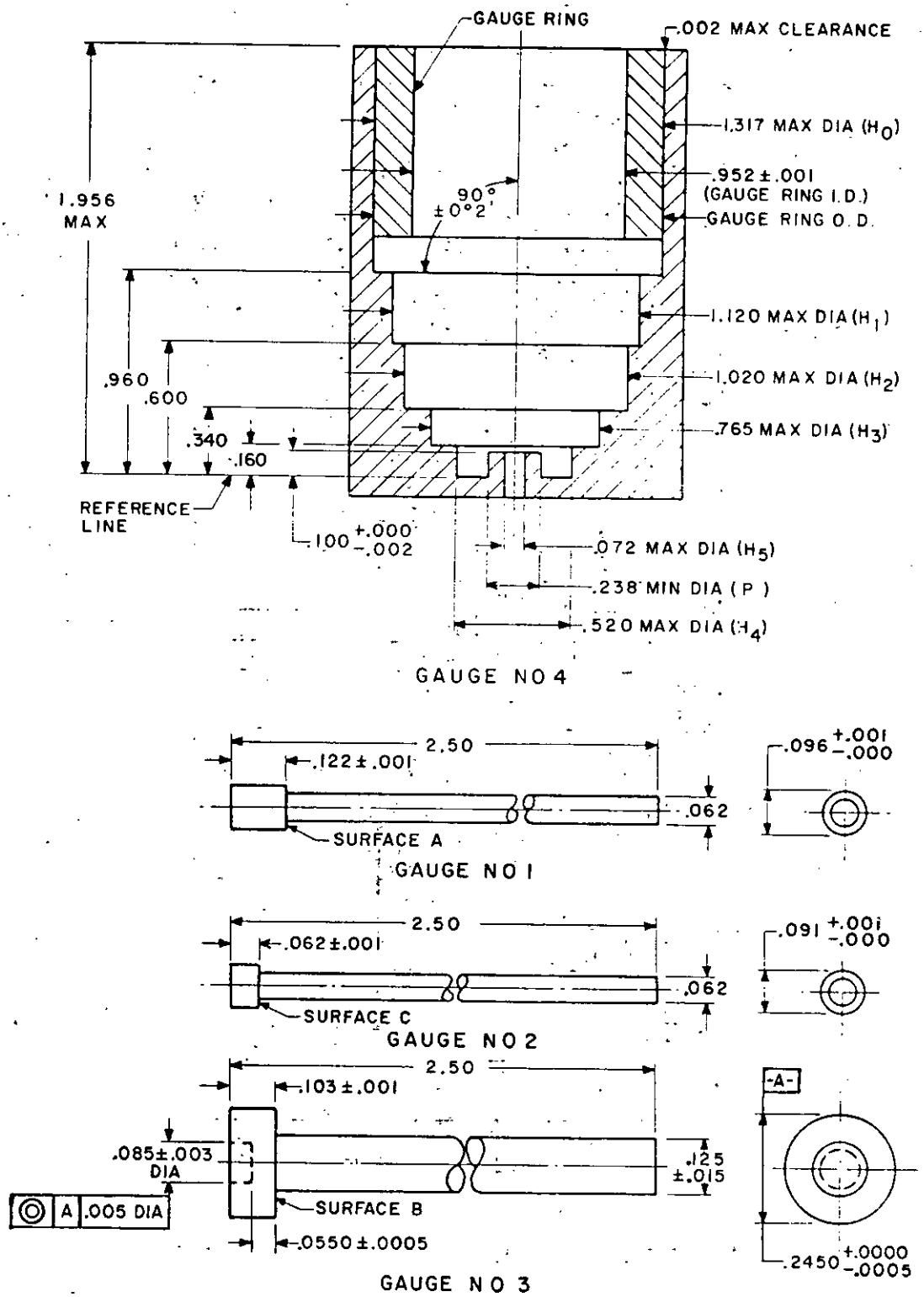


FIGURE 2. Outline drawing of electron tube type 7842 - Continued.

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INCHES	MM	INCHES	MM	INCHES	MM
.0000	.000	.085	2.16	.520	13.21
.000	.00	.091	2.31	.600	15.24
.0005	.013	.096	2.44	.765	19.43
.001	.03	.100	2.54	.952	24.18
.002	.05	.103	2.62	.960	24.38
.003	.08	.122	3.10	1.020	25.91
.005	.13	.125	3.18	1.120	28.45
.015	.38	.160	4.06	1.317	33.45
.0550	1.397	.238	6.05	1.956	49.68
.062	1.57	.2450	6.223	2.50	63.5
.072	1.83	.340	8.64		

NOTES:

a. With the cylindrical surfaces of the anode (cooler, or conduction cooling cylinder), anode terminal, screen grid terminal, control grid terminal, heater-cathode terminal, and heater terminal clean, smooth, and free of burrs, each tube will enter gage No. 4 to determine out-of-roundness, concentricity, and oversize. For Tube Type 7842 the gage ring shall enter between gage No. 4 and anode cap of tube until insert rests on anode flange of tube. The axes of the gage ring ID and gage ring OD are coincident within .001 inch (.03 mm). The axes of cylindrical holes H_0 through H_5 and the axis of post P shall be concentric within .001 inch (.03 mm). Proper seating is determined by failure of a .010 inch (.25 mm) thickness gage (1/8 inch wide) to enter between heater cathode terminal and bottom surface of H_4. A slot is provided to permit this measurement to be made.
b. This dimension to be checked by gage No. 1 entering this area. The edge of the grid No. 1 terminal shall be flush with or extend beyond the surface "A" on the gage.
c. This dimension to be checked by gage No. 2 entering this area. The edge of grid No. 2 terminal shall be flush with or exten d beyond the surface "C" on the gage.
d. This dimension to be checked by gage No. 3 entering this area. The edge of the heater terminal shall be flush with or extend beyond the surface "B" on the gage.
e. Dimension "M" applies over length "D" only.
f. Dimension "N" applies over length "F" only.
g. Dimension "P" applies over length "E" minus "G" only.
h. Dimension "R" applies over length "G" only.
j. On any one tube this dimension shall never be greater than "M".
k. On any one tube this dimension shall never be greater than "N".
m. These part 3 tests shall be performed every 6 calendar months.
n. Dimensions are in inches, unless otherwise specified.
p. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
q. This dimension checked by inserting a thickness gage of .050 inch (1.27 mm) minimum between the two fins.
r. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
s. The AQL for all dimensions listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

FIGURE 2. Outline drawing of electron tube types 7457 and 7842 - Continued.

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